Course Code & Name	BAI503 - Machine Learning	Credits	4
Name of the Course Coordinator	Prof. Sanjay M Belgaonkar	Sem	V
Question Bank			

Module - 1

Qn. No.	Sample Questions			
1.	Define a well-posed learning problem with an example.			
2.	List the major components of a learning system.			
3.	Explain the key characteristics of a well-posed learning problem.			
4.	Describe the primary perspectives and issues in machine learning.			
5.	Illustrate how a learning system can be designed for a spam email classification task.			
6.	Compare and contrast well-posed and ill-posed learning problems, with examples.			
7.	Critique the role of labelled data in ensuring a well-posed learning problem.			
8.	Define concept learning and explain its importance in machine learning.			
9.	What is the role of a hypothesis in concept learning?			
10.	Explain the general-to-specific ordering with an example.			
11.	Describe the purpose of the Candidate-Elimination Algorithm.			

12. Consider the following dataset for the concept **"EnjoySport"**, apply FIND – S algorithm to find the final hypothesis.

Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

13. By applying FIND – S algorithm determine the factors that lead to a positive hiring decision. The dataset has the following attributes:

Degree	Experience	Skills	Portfolio	Hired
Masters	5 years	Python	Good	Yes
Bachelors	4 years	Python	Good	No
Masters	6 years	Python	Good	Yes
Masters	5 years	Python	Excellent	Yes

14. Apply FIND – S algorithm to determine whether the weather conditions are suitable for a picnic based on attributes given below.

Outlook	Temperature	Humidity	Wind	Picnic
Sunny	Hot	Normal	Weak	Yes
Sunny	Hot	High	Weak	Yes
Overcast	Hot	High	Strong	Yes
Rain	Mild	Normal	Weak	No
Sunny	Mild	Normal	Strong	Yes

15. Demonstrate the use of the Candidate-Elimination Algorithm to narrow down hypothesis space for the following specific problem.

Outlook	Temperature	Humidity	Wind	PlayTennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes

16. Given the following training data for classifying fruits as 'Apple' or 'Not Apple', apply the Candidate Elimination Algorithm and show the changes in S and G after each example:

Example	Colour	Size	Texture	Apple
E1	Red	Small	Smooth	Yes
E2	Green	Large	Rough	No
E3	Red	Medium	Smooth	Yes
E4	Yellow	Large	Rough	No
E5	Red	Small	Rough	No

17. Compare the Candidate Elimination Algorithm with the FIND-S algorithm. How do they differ in terms of hypothesis search space and learning capability? Which algorithm would be preferable in cases with noisy data, and why?

18. Analyze the limitations of FIND-S in identifying the correct hypothesis.

19. Differentiate between FIND-S and the Candidate-Elimination Algorithm in terms of their approach to concept learning.

20. Evaluate the significance of inductive bias in concept learning. Support your argument with examples.

Module - 2

1. Explain the significance of supervised learning in machine learning.

2. Explain how model complexity affects overfitting and underfitting with diagrams.

3. Explain how k-Nearest Neighbours works for classification problems.

4. Describe the working of support vector machines for binary classification.

5. Define uncertainty estimation in the context of supervised learning.

6. The following dataset contains the number of hours studied (independent variable) and the marks scored in an exam (dependent variable) by 10 students. Build a simple linear regression model to predict the marks scored.
"Hours_Studied": [1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0]
"Marks_Scored": [35, 50, 60, 70, 80, 85, 90, 94, 97, 98]

7. Given the following confusion matrix, calculate accuracy, precision, recall, and F1 score.

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

8. A binary classifier has the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	30	20
Actual Negative	10	40

Calculate the precision and recall. Discuss how the classifier might improve precision at the expense of recall, and vice versa.

9. You have two models for predicting customer churn. Model A has precision = 0.6 and recall = 0.8, while Model B has precision = 0.75 and recall = 0.5. Analyze which model has a higher F1 score, and what does this imply about each model's performance?

10. Consider a fraud detection system where you evaluate 200 transactions, with 20 being fraudulent and 180 being legitimate. The model produces the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	**16**	**4**
Actual Negative	**30**	**150**

You are tasked with building a classifier to detect fraud. Why might recall be more important than precision in this case? Calculate the corresponding recall and precision.

11. Apply k – NN algorithm on the following dataset to classify the point (5,5).

Point	1	2	3	4	5	6
X	2	3	4	5	6	7
Y	4	3	6	7	4	5
Class	A	A	B	B	A	B

i. Choose k = 4 to check for a tie.
ii. Experiment on changing k to odd numbers like 3 & 5. Provide proper conclusion.

12. Build a decision tree to predict if a customer will buy a laptop based on the following dataset.

Price	High	High	Medium	Medium	Low	Low
Brand	Popular	Unpopular	Popular	Popular	Unpopular	Popular
RAM(GB)	16	8	8	4	4	8
Buy	Yes	No	Yes	Yes	No	Yes

13. Given the following dataset, identify whether it is suitable for classification or regression, and train a simple decision tree model:

Feature 1 (X)	Feature 2 (X)	Label (Y)
1.2	3.4	Class A
2.1	4.0	Class B

14. Build a logistic regression model on the given dataset and interpret the learned coefficients:

Hours Studied	Passed Exam (Yes/No)
2	No
5	Yes

15. Interpret the coefficients of a linear regression model used to predict house prices based on size and location.

16. Compare k-NN with logistic regression for a binary classification task in terms of performance and computational cost.

17. Explain how Naive Bayes handles categorical and numerical features in classification.

18. What is a decision tree? Explain its components. Describe how decision trees use entropy and information gain for splitting nodes.

19. Construct a decision tree to classify patients based on symptoms (e.g., fever, cough, fatigue) into "flu" or "cold." Make use of the following dataset.

Fever	Cough	Fatigue	Diagnosis
High	Severe	Present	Flu
Low	Mild	Absent	Cold
High	Severe	Present	Flu
Low	Mild	Absent	Cold
High	Mild	Present	Flu
Low	Severe	Absent	Cold

20. Explain how bagging and boosting differ in their approach to building models.

Module - 3

1. Define unsupervised learning and list two examples of its applications.

2.	Differentiate between clustering and dimensionality reduction in unsupervised learning.			
3.	Define feature scaling and describe its importance in machine learning.			
4.	Explain the differences between normalization and standardization with examples.			
5.	Analyze the impact of scaling features on clustering algorithms like k-means.			
6.	Describe the differences between Principal Component Analysis (PCA) and t-SNE.			
7.	Analyze how dimensionality reduction techniques can mitigate the curse of dimensionality.			
8.	Analyze how feature extraction differs from feature selection, with examples.			
9.	Compare density-based clustering (e.g., DBSCAN) with centroid-based clustering (e.g., k-means).			
10.	Evaluate the importance of choosing the right distance metric in clustering algorithms.			
11.	Analyze how data sparsity impacts the performance of unsupervised learning models.			
12.	Suppose you have a dataset with 1000 samples and 50 features. Model the application of PCA to reduce the dataset from 50 dimensions to 2 dimensions for visualization.			
13.	Given the covariance matrix: $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$ i. Construct & solve the matrix equation to find its eigenvalues and eigenvectors. ii. Identify the direction of the principal components.			
14.	Given the dataset below, reduce the dimensions from two to one by applying PCA. $\text{Data: } \begin{bmatrix} 2 & 4 \\ 0 & 0 \\ 1 & 1 \\ 3 & 3 \\ 5 & 5 \end{bmatrix}$			
15.	Compare the advantages and limitations of unsupervised learning compared to supervised learning.			
16.	Provide a solution to address the challenge of high-dimensional data in unsupervised learning using dimensionality reduction.			
17.	Evaluate the role of dimensionality reduction in improving computational efficiency.			
18.	Define clustering and list three clustering algorithms.			
19.	Explain the differences between hierarchical clustering and k-means clustering.			
20.	Evaluate the importance of choosing the right distance metric in clustering algorithms.			
21.	Apply the K-Means clustering algorithm to the following dataset with k = 2 and show the intermediate steps: (2,3), (3,4), (5,6), (8,8), (9,10).			
22.	Perform K-Means clustering with k=2. Compute centroids and assign points to clusters. Illustrate the results on a 2D plot. (1,1), (2,1), (1,2), (10,10), (10,12), (12,10)			
Module - 4				
1.	Define categorical variables. List two methods for handling them in machine learning models.			
2.	Explain the difference between one-hot encoding and label encoding with examples.			
3.	Apply one-hot encoding to the categorical variables in the following dataset and explain the resulting transformations. ID | Color | Size 1 | Red | Large 2 | Blue | Medium 3 | Green | Small 4 | Red | Medium			
4.	What is binning in feature engineering? Provide an example. Explain the difference between equal-width binning and equal-frequency binning with examples.			

5.	Compare the advantages and limitations of binning continuous variables in predictive modelling.			
6.	Explain how decision trees differ from linear models in their ability to capture non-linear relationships.			
7.	What are interaction features? Provide an example of their use in machine learning.			
8.	Explain how polynomial features can help capture non-linear relationships in data.			
9.	Discuss the challenges of overfitting when adding higher-degree polynomial features to a model.			
10.	Define univariate nonlinear transformations and provide two examples.			
11.	Explain the use of log transformations for skewed numerical features.			
12.	Compare the effects of log and square root transformations on model performance.			
13.	Define automatic feature selection and list two methods used for it.			
14.	Describe the differences between filter, wrapper, and embedded methods for feature selection.			
15.	Analyze how Recursive Feature Elimination (RFE) works and its limitations in high-dimensional datasets.			
16.	Encode the following categorical variable ["Low", "Medium", "High", "Medium", "Low"] using one-hot encoding and label encoding. Show the output.			
17.	Divide the feature Age = [18, 25, 35, 45, 55, 65] into three equal-width bins. Assign labels to each bin and show the resulting transformation.			
18.	A continuous variable "Income" is highly skewed. Apply equal-frequency binning and analyze how it can address the skewness in the dataset.			
19.	Fit a linear regression model to the dataset X = [[1], [2], [3], [4], [5]] and y = [2.1, 4.3, 6.5, 8.7, 10.9]. Calculate the slope and intercept of the regression line.			
20.	Use polynomial regression with degree 2 on the dataset X = [[1], [2], [3]] and y = [1.5, 4.2, 9.7]. Fit the model and evaluate its performance compared to linear regression.			
21.	Apply a log transformation to the variable Sales = [100, 200, 400, 800]. Explain how this transformation helps in handling skewed data.			
22.	Compare the effect of applying square root and log transformations on the variable "Income" = [1, 4, 16, 64, 256] and analyze their impact on variance stabilization.			
Module - 5				
1.	Define cross-validation and explain its purpose in machine learning.			
2.	Compare k-fold cross-validation with leave-one-out cross-validation.			
3.	Analyze how cross-validation mitigates the risk of overfitting when evaluating model performance.			
4.	What is grid search, and how is it used in hyperparameter tuning?			
5.	Describe the difference between grid search and random search. Which one is more efficient for high-dimensional hyperparameter spaces?			
6.	Discuss the computational challenges of grid search and propose strategies to address them.			
7.	Analyze the impact of hyperparameter tuning using grid search on a support vector machine's performance.			
8.	Define precision, recall, and F1-score. Provide their mathematical formulas.			
9.	Explain the importance of the ROC curve and AUC in evaluating classification models.			
10.	Analyze the difference between accuracy and F1-score in the context of an imbalanced dataset.			
11.	Evaluate the performance of a regression model using R^2, mean absolute error (MAE), and root mean square error (RMSE). Discuss the significance of each metric.			
12.	What is a scoring function, and why is it important in model evaluation?			
13.	Differentiate between scoring functions for regression and classification tasks with examples.			
14.	Analyze how scoring methods, such as log-loss and hinge loss, affect model optimization in classification tasks.			

15.	Evaluate the significance of using custom scoring metrics for domain-specific applications, such as medical diagnosis or fraud detection.			
16.	Explain how 5-fold cross-validation splits a dataset. Use the hypothetical dataset given below with 10 samples to illustrate your answer. Dataset: [S1, S2, S3, S4, S5, S6, S7, S8, S9, S10]			
17.	Differentiate between precision, recall, and F1-score. Explain with an example why F1-score is preferred in cases of imbalanced datasets.			
18.	A binary classifier produced the following confusion matrix: $\text{Confusion Matrix: } \begin{bmatrix} \text{TP} = 50 & \text{FP} = 10 \\ \text{FN} = 20 & \text{TN} = 70 \end{bmatrix}$ Calculate the precision, recall, F1-score, and accuracy for this model.			
19.	In fraud detection, false negatives are 5 times costlier than false positives. Propose and calculate a weighted scoring metric for a model with the following confusion matrix: TP = 30, FP = 15, FN = 10, TN = 45			
20.	A credit card fraud detection system must balance precision and recall. Explain how you would tune a classifier for this application using cross-validation, grid search, and a custom scoring metric.			